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EDITOR'S PICK

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Metabolic activity diffusion imaging (MADI): II. Noninvasive, high-resolution human brain mapping of sodium pump flux and cell metrics

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We introduce a new ^1H_2O magnetic resonance approach: metabolic activity diffusion imaging (MADI). Numerical diffusion-weighted imaging decay simulations characterized by the mean cellular water efflux (unidirectional) rate constant (k_{io}), mean cell volume (V), and cell number density (ρ) are produced from Monte Carlo random walks in virtual stochastically sized/shaped cell ensembles. Because of active steady-state trans-membrane water cycling (AWC), k_{io} reflects the cytolemmal Na^+, K^+ATPase (NKA) homeostatic cellular metabolic rate ($^cMR_{NKA}$). A digital 3D "library" contains thousands of simulated single diffusion-encoded (SDE) decays. Library entries match well with disparate, animal, and human experimental SDE decays. The V and ρ values are consistent with estimates from pertinent in vitro cytometric and ex vivo histopathological literature: in vivo V and ρ values were previously unavailable.

The library allows noniterative pixel-by-pixel experimental SDE decay library matchings that can be used to advantage. They yield proof-of-concept MADI parametric mappings of the awake, resting human brain. These reflect the tissue morphology seen in conventional MRI. While V is larger in gray matter (GM) than in white matter (WM), the reverse is true for ρ. Many brain structures have k_{io} values too large for

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Abbreviations used: See SI.IV.

current, invasive methods. For example, the median WM k_{io} is $22 s^{-1}$; likely reflecting mostly exchange within myelin. The $k_{io} \bullet V$ product map displays brain tissue $^{c}MR_{NKA}$ variation. The GM activity correlates, quantitatively and qualitatively, with the analogous resting-state brain ^{18}FDG-PET tissue glucose consumption rate ($^{t}MR_{glucose}$) map; but noninvasively, with higher spatial resolution, and no pharmacokinetic requirement. The cortex, thalamus, putamen, and caudate exhibit elevated metabolic activity. MADI accuracy and precision are assessed. The results are contextualized with literature overall homeostatic brain glucose consumption and ATP production/consumption measures. The MADI/PET results suggest different GM and WM metabolic pathways. Preliminary human prostate results are also presented.

KEYWORDS
brain, cell density, cell volume, high resolution, human, maps, metabolic activity, noninvasive

1 | INTRODUCTION

1.1 | MRI limitation

Water proton ($^{1}H_2O$) magnetic resonance imaging (MRI) has become ubiquitous in modern clinical medicine. It noninvasively maps tissue anatomy/morphology, even dynamically (e.g., cardiac motion, blood flow), with exquisite spatial resolution. Only computed tomography, with its attendant radiation dose, can compete. However, MRI has not been able to report metabolic activity, crucial in all biology. (Brain function is detected only indirectly, mainly via blood flow changes [functional MRI]). A handful of nuclear medicine and MR spectroscopic imaging techniques can determine metabolic activities, but with significantly poorer spatial resolution.[1] Addition of such metabolic capability could have a major impact on MRI. A list of acronyms and symbols is found in the supporting information (S.IV).

1.2 | The sodium pump; Na^+, K^+-ATPase (NKA)

The ubiquitous plasma membrane Na^+, K^+-ATPase (NKA) is arguably biology's most vital enzyme. This P-type ATPase stores metabolic energy (from ATP hydrolysis) in, and maintains the membrane potential caused by, respectively, the trans-membrane [Na^+] and [K^+] gradients (Figure 1). The [Na^+] gradient, in particular, drives many crucial, secondary (coupled) active membrane uptake transporters via Na^+ cotransport (generic enzymes III in Figure 1). Unfortunately, the very fast homeostatic NKA cellular metabolic rate ($^{c}MR_{NKA}$), which could approach 10^{10} [Na^+ and K^+] ions (cycled)/s/cell or 90 fmol (ATP [consumed])/s/cell, has never been measurable in vivo.[1] NKA substrates intracellular ATP (ATP_i) and Na^+ (Na_i^+), and extracellular K^+ (K_o^+), are rendered in red in Figure 1. The natural product $ouabain_o$ (also in red) is a specific, extracellular NKA inhibitor.

In recent years, however, the active trans-membrane water cycling (AWC) process has been discovered.[1–8] In AWC, x mole H_2O are cycled across the cell membrane for every mole ATP hydrolyzed by NKA, moving 3 mole Na^+ out and 2 mole K^+ in Figure 1. The H_2O passes through secondary active directional water cotransporting membrane symporters (reviewed in[9,10]), whose rates are thus driven by the rate-limiting $^{c}MR_{NKA}$. Water efflux and influx cotransporters are represented as generic enzymes I and IV, respectively, in Figure 1. These could also be aquaporins operating unidirectionally in response to an NKA-maintained thermodynamic driving force. Stoichiometric ratio, x, values for both individual cotransporters and overall AWC, are discussed below. There could be 10^{12} H_2O molecules actively cycled/s/cell.[1]

1.3 | NMR measurement of AWC

The homeostatic (steady-state) cellular H_2O efflux (unidirectional) rate constant, k_{io}, can be elaborated in the first-order differential chemical kinetic rate law for trans-membrane transport,

$$k_{io} = k_{io}(p) + k_{io}(a) = \left(\frac{A}{V}\right) P_W(p) + \left(\frac{x}{[H_2O_i]\langle V \rangle}\right) {}^{c}MR_{NKA}\,, \qquad (1)$$

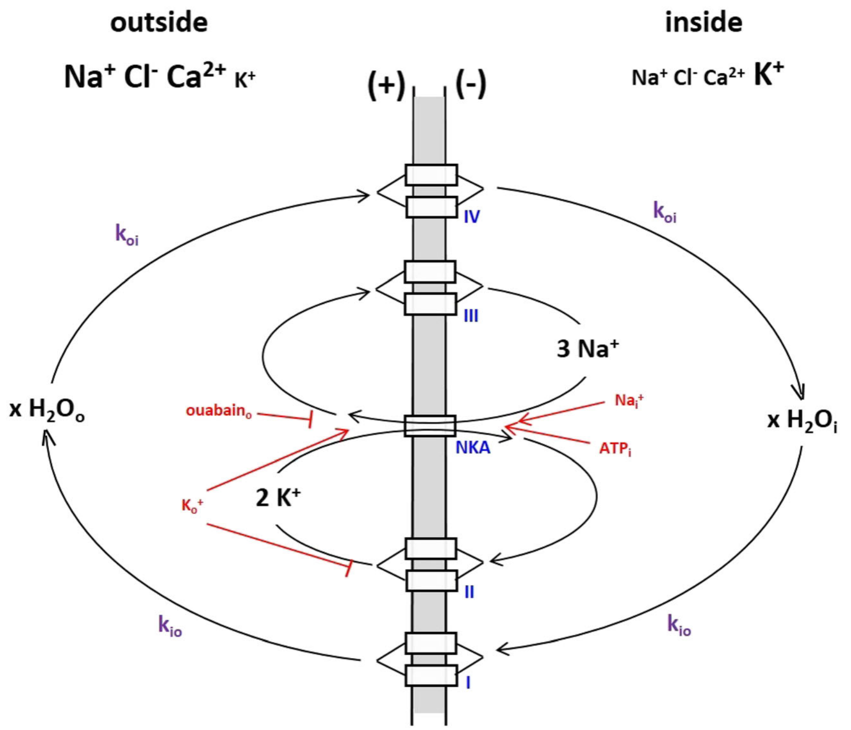

FIGURE 1 The general mechanism for steady-state, active trans-cytolemmal water cycling (AWC). The pseudo first-order (unidirectional) rate constants for water influx and efflux are k_{oi} and k_{io}, respectively. The AWC process is synchronized to, and driven by the Na^+, K^+ ATPase (NKA) activity. Potassium ion re-exit and sodium ion re-entrance transporters are indicated as generic enzymes II and III, respectively. Secondary active, directional water cotransporters for water efflux and influx[9,10] are indicated as generic enzymes I and IV, respectively. Three NKA substrates, extracellular K^+, and intracellular ATP and Na^+ (K_o^+, ATP_i, and Na_i^+), and the specific NKA inhibitor extracellular ouabain ($ouabain_o$), are indicated in red. At higher concentrations, K_o^+ becomes a K^+ efflux channel (a II enzyme) inhibitor. (A first approximation of this cartoon appeared in[1])

where $k_{io}(p)$ and $k_{io}(a)$ are the passive and metabolically active k_{io} contributions, (A/V) is the cell surface area/volume (claustrophobia) ratio, $P_W(p)$ is the cell membrane diffusive ("passive") water permeability coefficient, and $[H_2O_i]$ is the intracellular water concentration, near 42 M (see below). (In older literature, the k_{io} reciprocal—the mean intracellular H_2O molecule lifetime, τ_i—is often reported.) Historically, it was believed that the Equation (1) $k_{io}(p)$ term stood alone.[1]

It has proven possible to measure k_{io} with the tissue ^1H_2O magnetic resonance (MR) signal, the basis of by far the vast majority of in vivo MR studies. With current clinical techniques, human MRI can usually achieve slightly sub-mm spatial resolution. As indicated above, this is significantly better than any other in vivo imaging method capable of estimating enzyme activity, and for the first time offers the potential for high-resolution metabolic imaging.[1,5] In a recent review,[8] we tabulate 19 different demonstrations of k_{io} responsiveness to established, investigator-controlled metabolic changes. These independent results clearly demonstrate the presence, often dominance, of the second Equation (1) term. A few cogent examples follow.

Monitoring k_{io}, one observes Michaelis–Menten NKA signatures when titrating perfused rat hearts[3] and superfused rat brain slices[7] with low K_o^+ concentrations (1 to 8 mM) at constant $[Na_i^+]$ and $[ATP_i]$. At higher concentrations (> 10 mM), the K_o^+ role as K^+ channel (a II enzyme; Figure 1) inhibitor becomes dominant, the cell membrane potential is depolarized, and k_{io} decreases, almost perfectly matching synaptosome O_2 consumption.[7] Treatment with $[ouabain_o]$ reduces rat brain slice[6] and heart[3] k_{io}, as does reducing $[ATP_i]$ at constant $[Na_i^+]$ and $[K_o^+]$, or adding a specific NKA analog inhibitor, in yeast.[2] Superfusing brain slices with $tetrodotoxin_o$, a voltage-gated sodium channel (at least a III enzyme; Figure 1) inhibitor, or an inhibitor cocktail for excitatory, postsynaptic, glutamate neurotransmitter receptors, decreases k_{io}.[6,7] The k_{io} value increases with neuronal firing.[7] A doxorubicin-induced breast cancer cell k_{io} decrease correlates with decreased NKA-mediated Rb_o^+ influx.[11] A new k_{io} imaging biomarker could be extremely powerful.

1.4 | Brain k_{io}

However, the companion paper[12] reviews, and explains why, the most accurate k_{io} measurements are made only in such model systems, using paramagnetic Gd (III) chelate contrast agents (CAs). For a fundamental reason, measuring in vivo k_{io} accurately with CAs is quite difficult, if at all possible.[12] Furthermore, for the brain, where the normal blood–brain barrier (BBB) prevents CA extravasation, the situation is especially dire.

We did find CA can be used to measure the steady-state brain water capillary extravasation rate constant, k_{po}, in vivo.[5] k_{po} is an indirect measure of k_{io} within the neuro-glio-vascular unit (NGVU) due to metabolic coupling,[5] possibly via the communal K_o^+ NKA substrate.[10] Brain k_{po} maps are quite informative; white matter (WM) k_{po} decreases with aging,[13] in the normal-appearing (NA) multiple sclerosis (MS) brain,[5,14] in MS WM lesions,[5] in the choroid plexus with cognitive impairment,[15] and in brain glioblastoma (GBM).[5,16] Interestingly, the large, metabolically controlled k_{po} decreases, to $0.2s^{-1}$, in GBM tumors, while the tiny, passive CA extravasation rate constant, k_{pe}, increases, to 0.04 s^{-1}.[5,16] Water and CA utilize different capillary extravasation pathways, transcellular and paracellular, respectively.[5] Unfortunately, each k_{po} map requires a CA injection.[5]

To avoid CA, a diffusion-weighted (D-w) arterial spin label method has been developed and applied to MS.[17] However, this approach by itself confounds k_{po} with the blood volume fraction (v_b), as it measures the capillary water permeability coefficient (total) surface area product $P_W^{\dagger}S^{\dagger}$ ($= k_{po}v_b$). There could be cases where v_b changes oppose k_{po} changes (e.g., an $S^{\dagger}$ increase obscures a $P_W^{\dagger}$ decrease).

For these and CA environmental/safety reasons,[12] we have developed a completely noninvasive (CA-free) approach, metabolic activity diffusion imaging (MADI), based on D-w ^1H_2O imaging (DWI), to directly measure k_{io}, as well as two fundamental cytometric parameters: V, the mean cell volume (Equation (1)), and $\boldsymbol{\rho}$, the cell (number) density, also not previously accessible. The companion paper[12] details the conception, hypotheses, derivation, and execution of MADI modeling of DWI signal decays. We take advantage of the diffusion tensor (DT) magnitude, rather than its much more commonly used anisotropy (review[18]). MADI is complementary.

Because of the central NKA role in intermediary metabolism, k_{io} has diagnostic promise for any tissue or organ. Here, we further elaborate the MADI model and more extensively report its success in modeling experimental DWI data from disparate tissues. Finally, we show the first proof-of-concept MADI parametric maps of the in vivo awake, resting human brain. Metabolic activity mapping is demonstrated.

2 | METHODS

2.1 | MRI

Local Institutional Review Board-approved protocols were employed. Signed informed consent was obtained from all volunteers before study. T_1-weighted (T_1-w) and D-w images were acquired with a Prisma 3-T instrument (Siemens Medical Systems [SMS], Erlangen, Germany). Multislice DWI used a single-shot spin-echo (SE) echo-planar imaging (EPI) sequence. It employed Prisma 3-T implementation with bipolar (twice-refocused) gradients.[19] The whole-body coil set can produce G values up to at least 80 mT/m and slew rates up to 200 T/m/s. Three orthogonal pulsed field-gradient (PFG) directions were applied sequentially. Thus, DT trace-averaged data were obtained for analyses. For these, the vendor software numerically calculates $b = \gamma^2 M^2_{enc}\Delta_{ap}$, where M_{enc} is the PFG (zeroth) moment ($\sim\delta G$), and Δ_{ap} is the apparent diffusion time. The quantities b and Δ are defined and elaborated in the companion paper.[12]

Brain T_1-w imaging featured a 3D MP2RAGE (SMS) acquisition. A 20-channel head RF coil matrix (SMS) was employed. The two image contrasts were achieved with two TIs (2639/875 ms) with respective flip angles (4°/5°). The isotropic voxels had nominal dimensions—$(0.80\ \text{mm})^3$ (0.51 μL)—arising from a 240 x 256 mm^2 field-of-view (FOV) and TR/TE $=$ 5000/3.1 ms. With SMS software zero-filling, the effective resolution is $(1.60\ \text{mm})^3$.

Brain DWI data featured 16 evenly spaced b values ranging from 0 to 5000 s/mm^2, and incremented PFG strengths: $\Delta_{ap} = 36$ ms: a single diffusion-encoded (SDE)-CT acquisition. The same RF coil as for T_1-w (and in the same session) was used. The $\sim$31,000 voxels had nominal dimensions—(0.80 x 0.80 x 3.2) mm^3 (2.0 μL)—arising from a (192 x 256) mm^2 FOV, and TR/TE: 9500/92 ms. The nominal pixel area is identical to the T_1-w map, but with four times the thickness. With SMS software zero-filling, the effective resolution is $(1.60\ \text{mm})^2$. Thus the effective voxel volume is (1.60 x 1.60 x 3.2) 8.2 μL/voxel. Thus each voxel formally contains 8.2$\boldsymbol{\rho}$ cells. The total imaging time [TT] was 7.8 min.

aMRI parametric maps were visually smoothed with 1:4 2D bi-cubic interpolation. This does not change the effective in-plane resolution from $(1.60\ \text{mm})^2$. The $0.042 \bullet k_{io} \bullet V$ map was 2D Hamming-filtered. The effective in-plane resolution remains $(1.60\ \text{mm})^2$.

Prostate DWI data featured 15 unevenly spaced b values ranging from 0 to 5000 s/mm^2: only 0 s/mm^2 and the 10 b values greater than 150 s/mm^2 were used for analyses (to avoid potential blood flow effects). The PFG strengths were incremented: $\Delta_{ap} = 36$ ms. An endorectal RF receive coil was employed. The $\sim$4000 voxels had nominal dimensions ($(0.8\ \text{x}\ 0.8\ \text{x}\ 3\ \text{mm})^3$; 1.9 μL), arising from a $(240\ \text{mm})^2$ FOV, and TR/TE: 5100/91 ms. TT was 5.7 min.

2.2 | Prostate pathology

The subject underwent a standard 10-core prostate biopsy protocol under (exclusively) ultrasound guidance more than 6 weeks before the MRI imaging session. The biopsy cores were analyzed using standard histopathology procedures. The lesion in the MR image was assigned a Gleason (3 + 4) Score.

3 | RESULTS

3.1 | Simulated d^{rms} time-course and DWI decay libraries

We are assembling the virtual 3D libraries (dictionaries) of simulated diffusion displacement (d^{rms}) time-courses and SDE DWI decay curves.[12] The d^{rms} library is universal (canonical): there is only one. Because this requires high-performance computing,[12] library building is ongoing.

Examples of simulated SDE decays are shown in Figure 2, which presents families of simulated diffusion decays from 12 x 10^6 random walks (RWs) in different ensembles with the same k_{io}, ρ, and V values.[12] Values for $S(b)/S_0$ are calculated for 400 b values out to $b_{max} = 40{,}000$ s/mm^2, which has been reached with human brain data (Figure S1b; Tabzle S1). S_0 is the transverse ^1H_2O signal when b = 0, and $b_{SDE} = q^2t_D$, where q is the PFG wave number, and t_D is the diffusion time.[12] For our constant t_D (CT) simulations, q^2 is incremented. It would take averaging over many more RWs to avoid simulation artifacts manifest long before b reaches 40,000. These often appear as nonsmooth, or even nonmonotonic, decays. Thus we show simulations to only 5000 s/mm^2. The panels have ordinates that measure $\log_{10}(S(b)/S_0)$ (note the logarithmic scale), while the abscissae increment b. For Figure 2a, ρ was fixed at 1.2 x 10^6 cells/μL, and V at 0.66 pL (1 pL = 1000 μm^3). Decays for four finite k_{io} values, 12 to 71 s^{-1}, are shown. The decays are nonlinear in these semi-log plots; non-Gaussian behavior. Diffusivity increases with k_{io}. This is sensible: on average, water molecules can move further, and dephase more, the more permeable the cell membranes. Note the encouragingly good sensitivity to large k_{io}. For Figure 2b, k_{io} was fixed at 39 s^{-1} and V at 0.66 pL. Curves for five ρ values, 0 to 1.7 x 10^6 cells/μL, are shown. Diffusivity decreases with increasing ρ. This is also sensible: water molecules cannot move as far, the greater the "cellularity". The membrane "concentration", $\langle A\rangle\rho$, increases with increasing ρ.[12] For Figure 2c, k_{io} was fixed at 39 s^{-1} and ρ at 1.2 x 10^6 cells/μL. Decays for five V values, 0 to 0.74 pL,

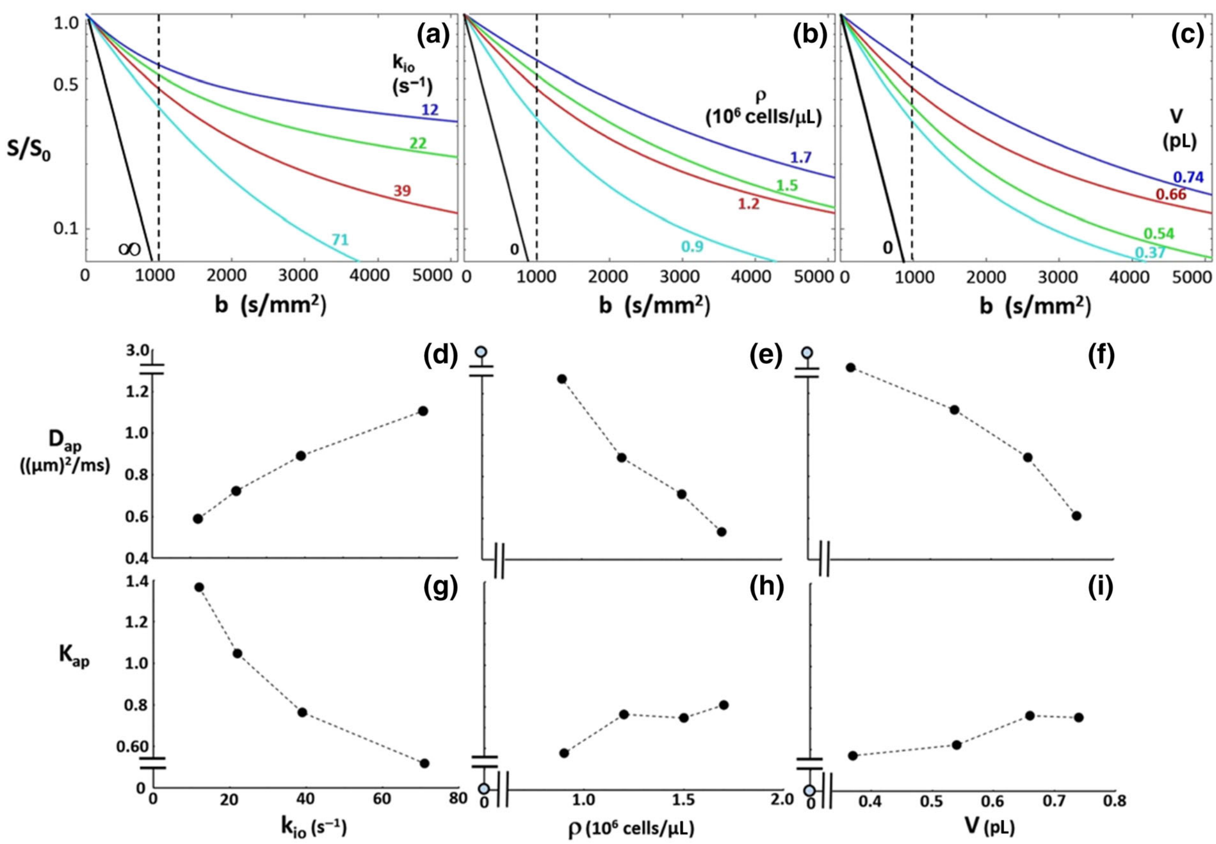

FIGURE 2 Semi-log plots of simulated single diffusion-encoded diffusion-weighted b-space decays. Each model parameter (k_{io}, ρ, V) is varied with the other two fixed. Increasing k_{io}, ρ, and V, respectively, causes diffusivity to (a) increase ($\rho = 1.2$ x 10^6; V = 0.66), (b) decrease ($k_{io} = 39$; V = 0.66), and (c) decrease ($k_{io} = 39$; $\rho = 1.2$ x 10^6). See[12] for simulation details. Panels (d), (e), and (f) show the k_{io}-, ρ-, and V-dependences of the apparent diffusion coefficient, D_{ap} (Equation 3), for the simulated panel (a), (b), and (c) curves, respectively. Increasing k_{io}, ρ, and V, respectively, causes D_{ap} to (d) increase, (e) decrease, and (f) decrease. Panels (g), (h), and (i) show the k_{io}-, ρ-, and V-dependences of the apparent kurtosis coefficient, K_{ap} (Equation 3), for the simulated panel (a), (b), and (c) curves, respectively. Only k_{io} < 30 s^{-1} causes K_{ap} to increase meaningfully above zero (Gaussian). The ρ– and V-dependences of D_{ap} ((e) and (f)) and K_{ap} ((h) and (i)) go smoothly toward the pure water D_{ap} ($\equiv D_0 = 3$ μm^2/ms) and K_{ap} ($\equiv 0$) values (light blue [open] circles). (Note the scale breaks.) For pure water, while $\rho = V \equiv 0$, k_{io} is undefined (but $+\infty$). The metabolic activity diffusion imaging (MADI) model is well behaved

are shown. As generally expected, diffusivity decreases with V increase, averaged over many cells. With V constant, the intracellular volume fraction v_i increases with ρ (Figure 2b); with ρ constant, v_i increases with V (Figure 2c). Thus the extracellular volume fraction v_o decreases in each case, causing "tortuosity" to increase. The decay for pure water ($k_{io} = \infty$, $\rho = 0$ cells/μL, and $V = 0$ pL) is linear, indicating a Gaussian diffusion process. The negative {ln(S(b)/S_0)/b} slope of this is $D_O = 3.0 \times 10^{-3}$ mm^2/s (37°C).

The relatively featureless SDE decay has two general properties important to us: (1) its absolute location in b-space, and (2) its degree of curvature. In clinical studies, a popular two-b-point approximation of the asymptotic [−ln(S(b)/S_0)/b] slope, the apparent diffusion coefficient (ADC), is commonly (but arbitrarily) evaluated with the S values at b = 0 and b = 1000 s/mm^2 (Figure 2a-c, vertical dashed lines). This is a poor, one-parameter (ADC) characterization, essentially of only property (1). Although also empirical, diffusion kurtosis is a natural two-parameter characterization of more of the decay. The kurtosis treatment arises from a cumulant expansion corresponding to a Taylor series expansion (Equation (2)[20]),

$$\frac{S(b)}{S(0)} = \left[e^{-D_{ap}}\right]^b \left\{ \left[e^{(D_{ap} D_{ap} K_{ap}/6)}\right]^b \right\}^b, \qquad (2)$$

where D_{ap} and K_{ap} are the apparent diffusion and kurtosis coefficients, respectively. Equation (3) is the more familiar logarithmic version of Equation 2.

$$ln\left\{\frac{S(b)}{S(0)}\right\} = -bD_{ap} + \frac{b^2 D_{ap}^2 K_{ap}}{6} \qquad (3)$$

The D_{ap} magnitude is a better estimation of the asymptotic slope than ADC, and a measure of property (1). The K_{ap} value measures the degree of non-Gaussianity; property (2). It can be positive, zero (signifying a Gaussian decay; linear semi-log plot), or negative.

We have fitted Equation (3) to the 10 different simulated decays in Figure 2a–c (the red decay is identical in each). The k_{io}-, ρ-, and V-dependences of D_{ap} are shown in Figure 2d–f, respectively, and of K_{ap} in Figure 2g–i. We see D_{ap} increases with increasing k_{io} and decreases with increasing ρ and V. Similar dependences to 2d and 2e are seen for ADC.[21] And it was shown some time ago that the DT trace, Tr[**D**], in WM, 0.7 μm^2/ms, is similar to that in GM, 0.8[22]: the fig. 3d-f D_{ap} values are consistent with these. The kurtosis, K_{ap}, is a measure of property (2), the semi-log decay curvature. All K_{ap} values here are positive, but increase meaningfully (>0.8) for only k_{io} less than 30 s^{-1} (Figure 2d). The pure water $D_{ap} = 3$ μm^2/ms and $K_{ap} = 0$ (Gaussian) values (pale blue [open] circles) are smoothly approached as ρ (Figure 2e,h) and V (Figure 2f,i) vary toward zero (note the scale breaks). The MADI model is well behaved in this sense. In pure water, k_{io} is undefined: in Equation (1), $\langle A/V \rangle \rightarrow +\infty$ when $V \rightarrow 0$.[12] Figure 2g shows k_{io} to be the parameter that is most influential on curvature. A smaller k_{io} value imparts greater curvature. In fact, for the smallest k_{io} values, Equation (3) cannot match the decays well (not shown): three parameters are necessary.

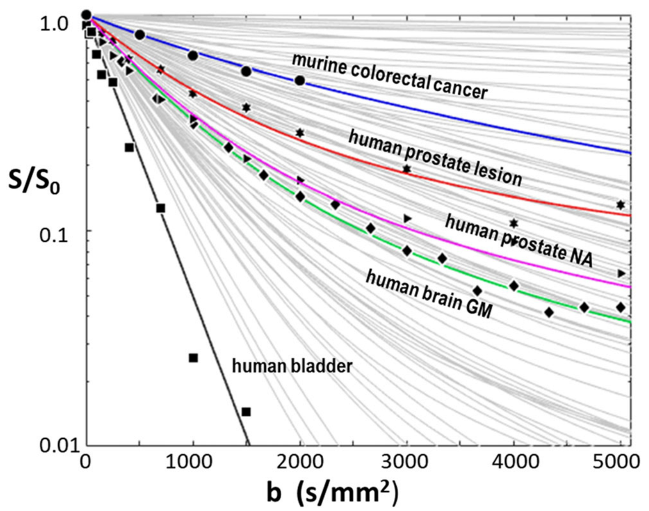

FIGURE 3 **b-Space Map.** Experimental 3-T data for human single voxel prostate lesion (stars), normal-appearing (NA) prostate (triangles), brain cortical gray matter (GM) (diamonds), and human bladder (squares) tissues (all this lab) and 4.7T data for a murine colorectal tumor (circles)[23] are shown. Different model k_{io}, ρ, and ⟨V⟩ parameter sets produce the colored simulated curves matching the disparate data. More importantly, the model parameters for each matching curve are in good agreement with measures from independent in vitro and ex vivo studies of such tissues (Table 1). The light gray curves comprise <0.5% of library-simulated b-space decays

In Figure 3, every 150th (only ~0.5%, for clarity) SDE-CT library entry is displayed as a light gray curve. These suggest our ultimate coverage of all possible decays, which must lie between the black line ($\rho = 0$) and the upper abscissa ($\rho \rightarrow \infty$). These can be considered the boundaries of "b-space". Every decay could have a unique combination of V, ρ, and k_{io} values. Currently, there are more than 14,000 entries: with 949 different V values spanning 0.01 to 206 pL, 186 unique ρ values from 4.4 x 10^3 to 58 x 10^6 cells/μL, 65 different k_{io} values spanning 0.0 to 130 s^{-1}, and 20 unique v_i values from 0.5 to 0.994. Thus there are 1300 (i.e., 65 x 20) different $\rho \bullet V$ hyperbolae upon which all library entries exist.[12] Some $\rho \bullet V$ pairs are impossible: $v_i = \rho \bullet V$ cannot exceed unity. Eventually, we expect to exceed 500,000 entries. Because of the good model behavior, we wish to investigate the utility of the current library by preliminary comparison of noiseless library decays with experimental data having varying levels of noise.

Thus Figure 3 also displays disparate experimental single voxel data, with noise, in order of increasing diffusivity, from murine xenograft colorectal (SW620) tumor (4.7 T, 31 nL voxels; circles),[23] and from human prostate lesion (stars) and NA (triangles) tissues, brain GM (diamonds) and bladder (in prostate FOV; squares) (details are available in the Methods). There are library entries that match each of the data decays reasonably well, and these are colored: colorectal cancer (blue), prostate lesion (red), NA prostate (violet), brain GM (green), and bladder (black; pure water). The simulated $\rho = 0$ line (Figure 2b, black; replotted in Figure 3) represents a Gaussian decay with $D_O = 0.003$ mm^2/s. This is also the case for an in vivo human cerebral ventricular ROI.[24] This is sensible, because the bladder and ventricular lumens are essentially acellular saline.

However, validation lies not just in good matching. There are many literature analytical three-parameter fittings of DWI data. Most are based on the assumption of an empirical (i.e., cannot be directly validated independently) bi-exponential decay, but, as we have found,[12] experimental decays result from continuous exponential distributions (i.e., an infinite number of exponentials). A MADI strength is that its parameters are not empirical, but real, quantities (with units), and have the potential to be compared with independent measurement for accuracy. In particular, the cytometric biomarkers V and ρ can be compared with quantities estimated from in vitro microscopy and ex vivo histopathology results. Table 1 shows the Figure 3 colored curve parameters (middle column) along with the most pertinent estimations from independent studies (right column). The general consistency in parameter ranges is encouraging. It is important to note, however, the relevant in vitro, and particularly histopathology, methods (cited) are necessarily ex vivo in nature. In the Discussion, we provide caution regarding comparison of in vivo and ex vivo parameters such as these.

TABLE 1 Simulated Figure 3 (single voxel/small ROI) DWI decays

Tissue	Simulation	Estimates from independent in vitro or ex vivo measures
parameter	**parameter**	**parameter**
murine colorectal cancer (SW620)	**blue curve**	
ρ (cells/μL)	950,000	1,002,000[23]
V (pL)	0.74	0.70[23]
k_{io} (s^{-1})	6.6	9.1 (168FARM breast cancer cells)[81]
human prostate lesion	**red curve**	
ρ (cells/μL)	1,200,000	660,000-centered broad distribution[39]
V (pL)	0.54	0.70[23]
k_{io} (s^{-1})	39	45.5 (4 T1 breast cancer cells)[81]
human NA prostate	**purple curve**	
ρ (cells/μL)	364,000	100,000-centered distribution[39]
V (pL)	1.2	0.8–2.5[23,48,49,82]
k_{io} (s^{-1})	40	too large for s-s-DCE-MRI[37]
human brain gray matter	**green curve**	
ρ (cells/μL)	74,000	see Discussion section
V (pL)	8.5	see Discussion section
k_{io} (s^{-1})	0.6	see Table 3
human bladder	**black line**	
ρ (cells/μL)	0	acellular saline
V (pL)	0	acellular saline
k_{io} (s^{-1})	undefined	undefined

Abbreviations: DWI, diffusion-weighted imaging; NA, normal-appearing; ROI, region of interest; s-s-DCE-MRI, shutter-speed dynamic contrast-enhanced MRI.

3.2 | In vivo MADI parameters

We do not use the intracellular volume fraction, v_i, as a variable parameter: it confounds the two more fundamental cytometric ρ and V parameters ($v_i = \rho \bullet V$). Also, its independent, global determination currently requires invasive indicator dilution techniques,[25,26] and its mapping requires invasive DCE-MRI: $v_i \cong 1 - v_e$, where v_e is the extracellular volume fraction (also v_o or ECV). Moreover, accurate ECV requires shutter-speed (s-s)-DCE-MRI, to account for water exchange kinetics,[27] and we have discussed its problems in the companion paper.[12]

The good agreement of the Figure 3 simulated decays with disparate experimental data spanning much of b-space, along with the simultaneous general consistency of their parameters with estimations from pertinent independent measurements (Table 1), is encouraging. This suggests the "forward problem"—the model prediction of experimental data—is in hand. The "inverse problem"—the uniqueness (precision) of experiment/model matching—is a separate question. For example, the Figure 3 purple prostate NA and green brain GM curves are rather close in b-space but have quite different parameter values (Table 1).

3.3 | Human brain parametric maps

The inverse problem is encountered in producing desired quantitative pixel-wise tissue parametric maps. In vivo cell membrane geometric complexity makes analytical solution of any tissue diffusion equation impossible. Our library-based RW approach numerically produces in silico DWI decay curves in parameter ranges suitable for in vivo tissues. Under the assumption that a small variation of any parameter results in only a small change in the slowly varying decay curve, we can extract the parameter set best matching in vivo data by least-squares comparison of an experimental decay with all library entries in a noniterative analysis.

As the SDE-CT library becomes more complete, we choose the Figure 3 human brain 3D DWI acquisition for proof-of-concept parametric mapping investigations. Figure 4 shows results for the awake, resting brain of a healthy 54-year-old male. Panel (a) presents an axial T_1-weighted image slice (inferior perspective: image left is subject right) at a level including the caudate (c), putamen (p), and thalamus (t) structures (indicated): T_1-w pixel intensity is proportional to voxel intracellular macromolecule content.[8]

The analogous DWI image slice comprises ~31,000 pixels. The b-space decays of 18,309 were individually least-squares compared with each of the library entries, that is, there were no constraints (except $v_i \leq 0.95$). Pixels clearly representing essentially acellular voxels (ventricular, subarachnoid, perivascular [Virchow–Robin] spaces, etc.), as seen in the T_2-w image (not shown), were not submitted for matching, and are colored black in Figure 4b-d. Recall: for pure water, $\rho = V = 0$, and k_{io} is undefined (becomes infinite) (Figure 2). For voxels containing acellular water

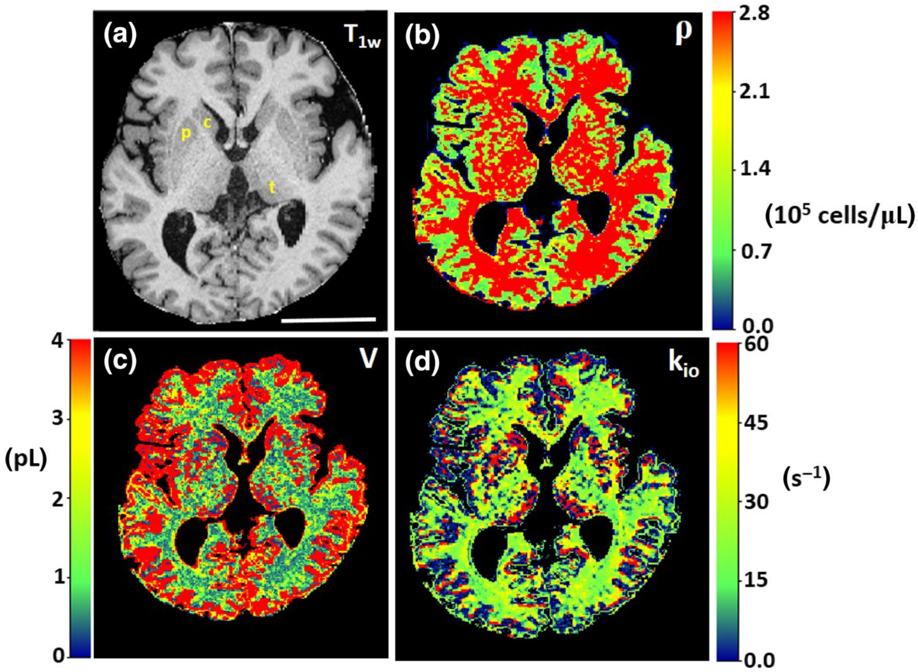

FIGURE 4 Axial in vivo normal, awake, resting human brain images (Figure 3 subject); inferior perspective (image left is anatomical right). (a) T_1-weighted image, (b) Cell density map, (c) Mean cell volume map, and (d) Mean k_{io} map. In the text, qualitative and quantitative map aspects are compared with estimates from literature in vitro and ex vivo measurements. Note, in particular, the large white matter k_{io} values. In the T_1-w image, the scale bar represents 5 cm, and brain substructures are indicated (c, caudate; p, putamen; t, thalamus)

portions, the ρ value will naturally decrease. However, if the v_i value ($\equiv \rho \bullet V$) does not decrease commensurately, this seems to cause artifactual V increases. Our library currently has no entries with v_i less than 0.5.

Panels (b-d) show, respectively, the MADI ρ, V, and k_{io} parametric maps obtained. The anatomy/morphology seen in the T_1-w image is reflected in the parametric maps, with good conspicuity. This indicates MADI analyses are discriminating tissues. In the Methods section, we report the parametric pixels have the same nominal areas as their T_1-w counterparts (but four times the thickness): the color maps have effective in-plane resolution $(1.60\ mm)^2$, with some spatial interpolation.

Figure 4 is the first mapping of in vivo ρ, V, and k_{io} achieved. It shows that ρ_{GM} is less than ρ_{WM} (panel b), while V_{GM} is more than V_{WM} (panel c). Interestingly, the median k_{io} (WM), $22s^{-1}$, is larger than the median k_{io} (cortical GM), $6.6\ s^{-1}$ (panel d): see below. It should be noted the WM k_{io} value is greater than considered accessible by s-s-DCE-MRI,[8] even if CA could pass the BBB. The GM/WM ρ and V trends are consistent with the ex vivo literature. There is greater cell nucleus number density in human WM than cortex, while cortex has a larger cell nucleus fraction of neurons,[28] which are larger than glia. Encouragingly, the v_i ($\rho \bullet V$) map (not shown) is relatively uniform near 0.8. We observe similar ρ, V, and k_{io} trends in preliminary studies of a second human subject and of the anesthetized rat and mouse brain.

We used the T_1-w image (Figure 4a) to create a mask segmenting[5,13] the MADI parametric maps into five distinct brain structures (cortical GM, thalamus, putamen, caudate, and WM). We then examined histograms of the matching returned pixel values for each of the three parameters in each of the five structures. As examples, Figure 5 shows the matching returned pixel V value histograms for cortical GM (a) and WM (b). There were 7205 pixels matched for cortex, and 8450 pixels for WM. Of course, the histogram distributions are discontinuous, because the library entries are discrete. Also, the distributions from these heterogeneous tissues are not Gaussian. (Estimations of V and ρ distributions from the independent in vitro and ex vivo literature are considered in the Discussion.) The cortical gray matter (cGM) V histogram (Figure 5a) is broader, with apparent maxima at least near 0.4 and 9 pL. On the other hand, the WM histogram (Figure 5b) is much narrower, with a maximum near 0.6 pL and a small, broad shoulder near 3 pL. These are as expected because human cortex is known to have a greater cell nucleus fraction of larger neurons, while WM has a much larger fraction of smaller non-neuron cell nuclei.[28] However, we must also consider the nature of the library parametrization.

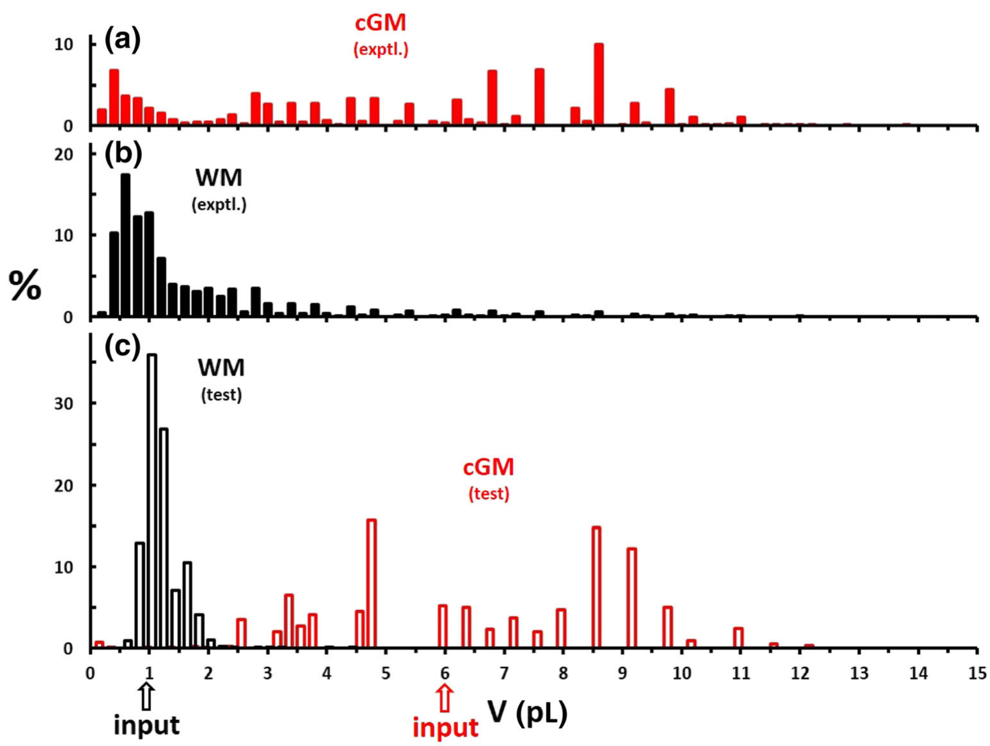

FIGURE 5 The voxel-wise histograms for V values returned from single diffusion-encoded (SDE)-CT library-matchings of the Figure 4 brain (a) cortical gray matter (cGM) (7205 voxels) and (b) white matter (WM) (8450 voxels) experimental decays. Statistical analyses of these discontinuous distributions are discussed in the text, and numerical results are given in Table 2 for these and the other brain regions. (c) Results of parameter uncertainty tests. Library decays representative of cortex ($\rho = 1.3 \times 10^5$ cells/μL, V = 6.0 pL, $k_{io} = 6.6\ s^{-1}$) and of WM ($\rho = 6.9 \times 10^5$ cells/μL, V = 0.9 pL, and $k_{io} = 22s^{-1}$), each with noise added randomly 50,000 times (S/N = 50), were submitted to the SDE library. The V values returned from the 50,000 matchings are shown for the cGM (red) and WM (black) tests, with their input V values indicated (up arrows). The up arrows apply to only panel (c). The bin size (bar width) is 0.2 pL. The significance of these results is discussed in the text. Note the generally good discrimination of GM from WM

3.4 | LIBRARY MATCHING OF EXPERIMENTAL DATA

Most multiparametric model/data fittings are analytical, that is, the parameters appear in a model equation of closed form. Thus, commonly the parameters are considered continuous variables. In most data fittings, the parameter values are varied (rather continuously) iteratively, and often independently of each other (e.g., (89), SI). These minimize the cost function with, say, fastest gradient descent. However, our library has only discrete ρ, V, and k_{io} parameter magnitudes, and the library entries consist of ρ, V, and k_{io} value sets. We cannot vary the three parameters absolutely independently of each other. The ~14,000 entries must lie on the library's 1300 $\rho \bullet V$ hyperbolae (see above). The parameters and parameter sets are discontinuous: the libraries have a granular nature.

An advantage of discrete libraries is they may mitigate the "ill-conditioned" inverse problem. A "brute force" approach—direct least squares comparison of each voxel decay with all library entries in a noniterative manner[29]—can avoid the specific pitfalls of local error surface minima to which gradient descent solvers are prone. Our library matching finds the global minimum readily, even if it is quite broad, as well as indicates local minima (see below).

Furthermore, the physical quantities represented by the V and k_{io} parameters are not necessarily continuous themselves. These are cell-based properties. Although there are 10^4–10^6 cells per voxel (recall: V and k_{io} are really $\langle V \rangle$ and $\langle k_{io} \rangle$, the voxel-averaged quantities), there are only finite numbers of cell types in tissues. Surely there are continuous variations of V and k_{io} for a given cell type, but these might be small compared with the values for different cell types.

For both of these reasons, we consider returned parameter distributions such as seen in Figure 5 discontinuous. They are complicated convolutions of the intrinsic parameter discontinuities and the tissue physical property discontinuities. These issues are manifest in the parameter uncertainty (or precision); the inverse problem.

In Table 2 we list the medians and ranges, of the ρ, V, and k_{io} parameter values returned for the voxels of each of the five segmented Figure 4 tissue types. For the caudate, we report only the right nucleus due to ventricular confounding of the left caudate nucleus (image right) in this image slice (Figure 4a): thus, this gives a small voxel count (Table 2). It is interesting to note the GM tissue (cortex, thalamus, putamen, and caudate) parameter ranges are larger than those of WM tissue, and they bracket the WM median values. This suggests tissue heterogeneity could contribute to the parameter distributions, and that it is greater in GM. This can be particularly appreciated in the k_{io} parametric map (Figure 4d). Although the median WM value (yellow-green) is greater than the median GM value, the spread of k_{io} values (red to dark) in GM is greater. Red GM k_{io} pixels are often seen near WM borders. This is possibly significant: the cortex is well known to have a cellularly heterogeneous, multiple annular layered structure.[30] None of the distributions in Table 2 exhibit a normal (Gaussian) shape.

We tested the intrinsic uncertainty (precision) of our library matching in the following way. We selected single library entries with parameter values near the Figure 4 cortex and WM median values (Table 2). These were: for cortex, $\rho = 1.3 \times 10^5$ cells/μL, V = 6.0 pL, and $k_{io} = 6.6\ s^{-1}$;

TABLE 2 Figure 4 segmented tissue voxel distribution histogram median, voxel-averaged MADI parameter values

Tissue	Symbol	Voxels	ρ[a] (10^5 cells/μL)	Range[b]	$\langle V \rangle$[a] (pL/cell)	Range[b]	$\langle k_{io} \rangle$[a] (s^{-1})	Range[b]	$\langle 0.042 \bullet k_{io} \bullet V \rangle$[a,c] (nmol[$H_2O_i$]/s/cell)	Range[b]
GM										
cortex	cGM	7205	1.1	1–16	6.0	0.5–10	6.6	3–72	1.0	0.21–2.5
thalamus	t	1518	6.2	2–16	1.2	0.5–9	22.0	3–72	0.95	0.21–1.9
putamen	p	907	2.9	2–26	2.6	0.5–9	17	3–72	0.89	0.21–1.9
caudate[d]	c	193	1.1	1–24	0.72	0.5–9	40	3–96	0.89	0.21–1.5
WM	WM	8450	6.9	2–16	0.91	0.5–4	22.0	12–42	0.83	0.42–1.5
Precision tests										
cortex	cGM									
test input		1	1.1		6.0		6.6		1.7	
matchings		50,000	1.6	1–3	4.7	4–9	11	3–15	1.5	0.5–2.5
WM	WM									
test input		1	6.9		0.91		22		0.84	
matchings		50,000	5.2	5–9	1.2	0.8–1.3	22	18–24	1.0	0.8–1.2

Abbreviations: cGM, cortical gray matter; FWHM, full-width-half-maximum; GM, gray matter; MADI, metabolic activity diffusion imaging; WM, white matter.

[a]median, voxel-averaged (⟨ ⟩) value.

[b]~FWHM of heterogeneous or discontinuous distribution ~70% CI; would span ~(±SD) for a normal, continuous distribution.

[c]$\langle 0.042 \bullet k_{io} \bullet V \rangle \neq 0.042 \langle k_{io} \rangle \langle V \rangle$.

[d]right caudate.

and for WM, $\rho = 6.9 \times 10^5$ cells/μL, V = 0.9 pL, and $k_{io} = 22s^{-1}$. To these noiseless library b-space decays, we added noise (to give S/N = 50) randomly 50,000 different times. Then we submitted the 100,000 noisy library decays back for library matchings, just as we did for experimental voxel decays. Histograms for the cGM and WM tests are shown in Figure 5c. The test median values and returned parameter value ranges are listed in Table 2, along with the test inputs.

These test results may illustrate both the weakness and the strength of noniterative library matching. The test cGM histogram (open red bars, Figure 5c) is very broad: almost as broad as the experimental cGM histogram (filled red bars, Figure 5a). On the other hand, the test WM histogram (open black bars, Figure 5c) is considerably narrower than the entire experimental WM histogram (filled black bars, Figure 5b). This represents a reasonable uncertainty (precision), but indicates one cannot tell if the breadth of the experimental WM peak near 0.6 pL reflects library V heterogeneity, tissue V heterogeneity, or both. Some of the WM tail to large V (Figure 5b) most likely does represent a few large tissue V values.

Why should the test WM distribution be so different from that of the cGM distribution? Because of the way the library was populated, the density of its different V values increases with decreasing V magnitude. Below 1 to 2 pL (the WM neighborhood), the V spacing is ~0.05 pL and decreasing. Above 8 to 9 pL (the cGM neighborhood), the V spacing is ~0.2 pL and increasing. One can see this reflected in the sparsity of bars at larger V values (Figure 5). Apparently, library matching precision is very sensitive to the density of available library entries.

Thus, the breadth of the test cGM histogram (Figure 1c) certainly reflects the library granularity. With the current library, one cannot tell whether the experimental cGM histogram (Figure 5a), or the parametric maps (Figure 4b–d), reflect the well-known,[30] and important, cGM cellular heterogeneity. Similar considerations obtain for the k_{io} and ρ histograms (not shown).

Interestingly, however, the test cGM histogram (red, open) is relatively symmetric about the input V value (up arrow), and the test WM histogram (black, open) has an almost normal distribution centered near the input V value (the returned median and input values are listed in Table 2). This suggests minimal systematic error, and that the results are reflecting the global minima (SI.I.), but there may be two local minima for cGM, near 4.8 and 8.6 pL (Figure 5c). The library entries "hit" in the cGM test can reside on different v_i hyperbolae near that of the input v_i (0.76), so not all V values on the $v_i = 0.76$ hyperbola are hit. Whatever the case, it is quite clear why MADI discriminates GM and WM tissues well (Figure 4c). Furthermore, this analysis indicates a clear path to improve MADI precision (see below).

For our current analyses, we use relatively simple metrics useful for discontinuous variables; for (a) magnitudes and median values, and for (b) ranges, estimates of the histogram full-width-at-half-maximum (FWHM), which corresponds to the ~70% confidence interval [CI], the span of the histogram bars accounting for ~70% of the hits would correspond to approximately ±SD for a normal, continuous distribution. As we will see, the parameter median values, along with the parametric maps, as well as careful consideration of pertinent in vitro and ex vivo literature and NMR principles, suggest reasonable MADI accuracy.

4 | DISCUSSION

4.1 | Relative parameter quantification

In Table 2, the cGM and WM (k_{io}, V) medians are 6.6 s^{-1}, 6.0 pLcell and 22/s^{-1}, 0.91 pL/cell, respectively. Consider the k_{io} contributions in Equation (1). The passive $k_{io}(p)$ term is $(A/V)P_W(p)$. We can estimate the $k_{io}(p)$ magnitude as follows. In deliberately ATP_i-depleted yeast, $k_{io} = 1.2$ (±0.3) s^{-1}.[2] Those yeast cells were nearly perfect spheres with V = 42 fL (0.042 pL). For a sphere, (A/V) = 1/(3r), where r is the radius; 2.16 μm for a 42 fL sphere. This gives $P_W(p) = 5.5$ μm s^{-1}. Assuming that to be the passive permeability coefficient for a lipid bilayer cell membrane, we calculate $k_{io}(p) = 0.33$ and 0.24 s^{-1}, respectively, for 2.0 and 5.5 pL spheres. Thus it is a reasonable approximation to neglect the Equation (1) $k_{io}(p)$ contribution, and to take $k_{io} = (1/[H_2O_i])(1/V)(x)^cMR_{NKA}$.

One could ask: Is the respective k_{io} difference, 6.6 and 22s^{-1}, for cGM and WM, respectively, due simply to the V difference for these two tissues? We can address this with the equality $[H_2O_i]•V•k_{io} = x•^cMR_{NKA}$. If the $[H_2O_i]•V•k_{io}$ product was identical for cGM and WM, then $x•^cMR_{NKA}$ would be the same in the two tissues. We take the intracellular water concentration $[H_2O_i]$ to be 42 M = 0.042 nmol(H_2O_i)/pL.[31] From Table 2, we see $0.042•V•k_{io} = 1.0$ and 0.83 nmol(H_2O_i)/s/cell for cGM and WM, respectively. The WM k_{io} is not as large as its small V would dictate: if that was the only factor. Thus, the $x•^cMR_{NKA}$ value is smaller for WM than for cGM. Below, we cite literature reports that both WM ATP_i production and total consumption rates are also smaller than in cGM. Furthermore, Table 2 gives the larger median $0.042•V•k_{io}$ values as 0.95, 0.89, and 0.89 nmol(H_2O_i)/s/cell for the internal GM thalamus, putamen, and right caudate structures, respectively.

With what can we compare these novel AWC ($x•^cMR_{NKA}$) measures? Figure 6a shows a scatterplot of Table 2 voxel-averaged, tissue-segmented median $x•^cMR_{NKA}$ values (vertical axis) versus $^tMR_{glu}$ values, in pmol (glucose)/s/μL (tissue), from the first quantitative, 30-min, pharmacokinetic ^{18}FDG-PET human brain study.[32] A dashed trend-line shows a strong correlation between these quite independent measurements of GM metabolic activity. Although $^cMR_{NKA}$ and $^tMR_{glu}$ measure the kinetics of different enzymatic processes, and represent cellular versus tissue properties (the latter confounded with ρ), it is reasonable that in tissues where glucose is metabolized faster, NKA ATP consumption is also faster. The main role of glucose metabolism is ATP production, and the brain is in overall homeostasis.

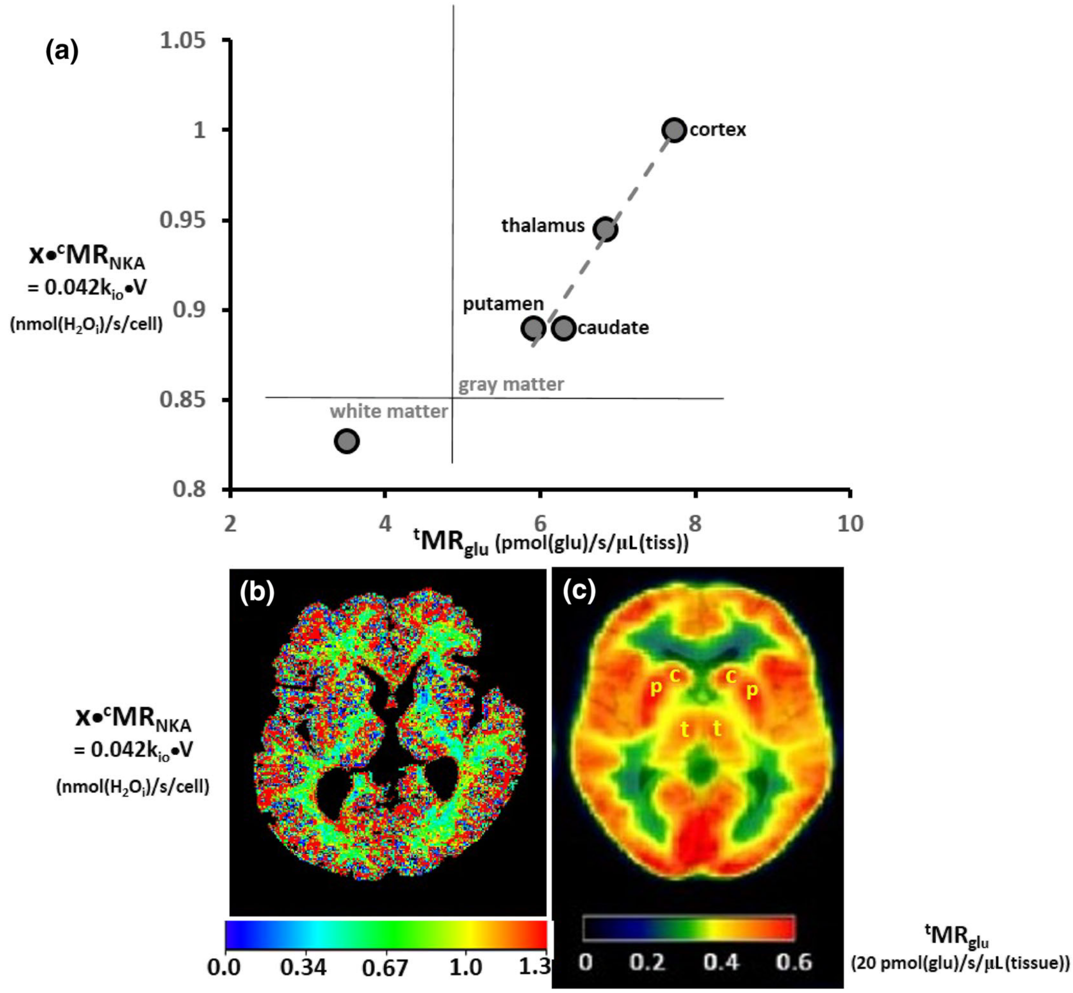

FIGURE 6 Metabolic activity diffusion imaging (MADI) correlation with PET. (a) Scatterplot of Table 2 segmented, voxel-averaged, median $[H_2O_i]$•V•k_{io} values (ordinate) versus ^{18}FDG-PET ⟨ROI⟩ $^{t}MR_{glu}$ values from[32] (abscissa). A dashed trend-line is shown for gray matter (GM) tissues. (b) $^{c}MR_{AWC}$ ($[H_2O_i]$•V•k_{io}) map ((1.60 mm)^2 in-plane resolution; effective voxel volume, 8.2 μL). $[H_2O_i]$ = 0.042 nmol/pL. (c) Quantitative ^{18}FDG-PET map ((2.50 mm)^2 in-plane resolution; effective voxel, 15.6 μL) modified from.[34] The rate of NKA-induced trans-membrane water cycling (nmol $[H_2O_i]$/s/cell) correlates with the glucose consumption rate (pmol [glucose]/s/μL [tissue]) both (a) quantitatively and (b, c) qualitatively. Both modalities agree white matter (WM) metabolic activity is less than that of GM. The WM point is in a different quadrant of panel (a). The noninvasive MADI map (b) is similar to the well-known pharmacokinetic ^{18}FDG-PET map (c)

The correlation is particularly interesting when one considers that there are alternative cell membrane glucose importer families (SGLT and GLUT) that could be differently distributed in tissues. These families contribute to both AWC and to ^{18}FDG-PET differently. SGLT has an individual mean stoichiometry of x' = 235 mol (H_2O) imported/mol (glucose) imported, while GLUT has only x' = 40.[9] SGLT does not transport FDG while GLUT does.[33] The correlation with PET supports the hypothesis that k_{io} is related to $^{c}MR_{NKA}$, and that the overall AWC stoichiometric coefficient x (nmol $[H_2O_i]$ cycled/nmol $[ATP_i]$ consumed) is relatively invariant between brain GM tissues, and that one glucose transporter dominates in healthy GM. It also illustrates a virtue of determining V, which cannot be accessed with the tracer or s-s pharmacokinetic paradigms. Both the MADI and ^{18}FDG-PET modalities agree WM activity is smaller than that in GM: the WM point appears in a different Figure 6a quadrant. That it does not fall on an extension of the dashed line suggests that x differs for GM and WM, the glucose transporters differ for GM and WM, or both. These results underscore the basic premise that the tissue $^{1}H_2O$ DWI signal carries metabolic activity information.

A more stringent test is to consider not median, but pixel-by-pixel, k_{io}•V products. Figure 6b maps these 0.042•k_{io}•V values. They are not uniform in the human brain (unlike the ρ•V map). The cortex, putamen, thalamus, and caudate regions are reddish—indicating large x•$^{c}MR_{NKA}$ values—high NKA ATP_i consumption. In panel (c), we reproduce a recent quantitative, axial ^{18}FDG-PET image (from a slightly different plane) of the awake, resting human brain.[34] The intensities are measures of the tissue metabolic rate of glucose consumption, $^{t}MR_{glu}$ (20 pmol [glucose] consumed/s/μL [tissue]), and are in good quantitative agreement with the first study[32] (Figure 6a). Qualitative similarities between the aMRI and PET images are clear. The effective in-plane spatial resolution of the AWC (x•$^{c}MR_{NKA}$]) map (Figure 6b) is (1.60 mm)^2; effective voxel volume, 8.2 μL, while that of the PET image (Figure 6c) is (2.50 mm)^2; effective voxel, 15.6 μL; two times larger. Greater apparent GM heterogeneity is again visible in the MADI map (Figure 6b), but not resolved in the PET map (Figure 6c) due to its lower spatial resolution. This gives the PET image an apparent smoother appearance.

$x \bullet {}^cMR_{NKA}$ mapping offers several advantages, its (a) improved resolution, (b) relatively lower cost, (c) noninvasiveness (no radiation dose or injection), and (d) nonpharmacokinetic nature. It can be repeated relatively quickly, and essentially endlessly. The MADI map (Figure 6b) suggests heterogeneity of metabolic activity that may be missed in the PET image (Figure 6c). We reported preliminary observations of the latter phenomenon in breast tumors.[1] As detailed above, however, we cannot tell if the apparent MADI map GM metabolic activity heterogeneity (Figure 6b) reflects true tissue heterogeneity, but ^{18}FDG-PET cannot resolve it either (Figure 6c). With an enhanced library, MADI might be able to do this. If so, there are possibilities for observing resting-state functional changes and/or stimulus responses. The k_{io} value increases with neuronal firing.[7] Brain ^{18}FDG-PET is sensitive to aging[35] as well as many neurological pathologies.[36] We might expect the same for MADI brain mapping. It seems to have potential to provide much neuroimaging richness. We discuss approaches for MADI refinement below.

From Table 1, we estimate the prostate lesion and NA voxel $0.042 \bullet k_{io} \bullet V$ values to be 0.88 and 2.0 nmol (H_2O_i) cycled/s/cell, respectively (although $\langle k_{io}\rangle\langle V\rangle \neq \langle k_{io}V\rangle$). The lesion $x \bullet {}^cMR_{NKA}$ is similar to those in the brain (Table 2), and the NA value is doubled. A preliminary s-s DCE-MRI prostate cancer mapping suggested that k_{io} was generally smaller in the tumor than in NA tissue.[37] Because the lesion V is also smaller than the NA prostate V (as expected), this would indicate reduced $x \bullet {}^cMR_{NKA}$ in the tumor. However, this must be further investigated. The lesion conspicuity in the in vivo ρ map[38] also has considerable diagnostic potential. It is matched only in sophisticated ex vivo cellularity maps.[39] However, the treatment utility of the latter is mostly nullified by the necessity for prostatectomy.

4.2 | Absolute quantification: independent aMRI parameter estimation

MADI is not an empirical approach: its parameters (having units) can potentially be compared with independent, absolute measurements. This could also provide a stringent test. However, these fundamental biomarkers have not before been reliably accessible in vivo. The difficulties with extant brain in vivo k_{io} reports were discussed in the MADI I. Introduction,[12] and provide the motivation for MADI. We will have more to say about this below.

We initially thought we could turn to classic 2D and/or 3D ("stereological") histopathology to assess the accuracy of our V and ρ values. But all relevant histopathology methods are necessarily ex vivo in nature. And, total tissue water content and intercompartmental water distribution, along with homeostatic trans-cytolemmal water exchange, are exquisitely sensitive to the living condition. Furthermore, these properties are effectively maintained by the ongoing activity of the Na^+/K^+ pump (NKA) itself.[40] Specific (ouabain) NKA inhibition in superfused brain slices causes essentially instant cell swelling,[6,41] that is, V increases due to net extracellular to intracellular compartmental water transfer. Sudden death also stops NKA activity. We reviewed many similar findings in.[12]

After excision, which itself can significantly alter tissue hydration,[42] tissue is chemically fixed before histopathology, and this can cause further such changes. For example, a fixation procedure has been reported to reduce rat brain tissue volume by ~33%,[40] due to water content loss. A net water content change alters the value of ρ. Even if excision has not already quenched NKA activity, some molecular process changes following fixation-induced NKA cessation have been itemized.[40,43] The fixing solution tonicity can alter V dramatically, nearly ±100% for some cells in the extreme (hypotonic/hypertonic solution swells/shrinks cells).[40] Interestingly, the optimal tonicity varies for different tissues.[40]

Therefore: the (1) organism death, (2) tissue excision, and (3) tissue fixation steps required in all histopathology can cause water redistribution, as well as likely water content change. Our in vivo cytometric parameters ρ and V (as well as k_{io}) are intimately involved with tissue water. In the final analysis, for truly in vivo tissues, none of the MADI parameters can be directly determined by any other method. There are no actual "gold standards" to validate parameter accuracies. Nonetheless, we seek the most pertinent in vitro and ex vivo and model measurements from which to make estimates.

4.2.1 | Mean cell volume (V)

As stated above, V is particularly sensitive to ex vivo tissue handling (death/excision/fixing). The incubation of fresh brain slices is an ex vivo approach not necessarily requiring fixing[41]: with proper culturing and then superfusion, such "organotypic" cortical slices can exhibit neuronal firing.[6,7] Such preparations are readily amenable to optical microscopy: fluorescent dye advances have made this particularly elegant. Estimations of MADI parameter values from analogous studies should be especially valuable.

Neural cells have highly dendritic structures.[30,44] However, straightforward geometrical analyses of quantitative fluorescence microscopic images of individual cells in incubated brain slices can yield realistic V estimates. In a comprehensive analysis, Nedergaard and colleagues showed human neural cells are larger than their rodent counterparts.[30] We carefully examined high quality immuno-fluorescent images of a human astrocyte (fig. 4b,[30]), found mostly in GM, and an oligodendrocyte[44] from WM. Conservatively large estimates of the volumes of component parts (processes or neurites [dendrites], along with the cell body [soma]) were made. The cell body is no larger than 0.5 pL (1 pL = 10^3 μm^3), and large process sections are no larger than 0.05 pL. If we add the cell body V to a reasonable estimate of the Vs of 100 processes, we obtain a total V of no more than 5 pL.

In human cortex, 42% of the cell nuclei are neuronal.[28] Our cGM histogram (Figure 5a) has a median value of 6 pL and a maximum near 8 pL. Although the Figure 5a distribution is likely dominated by library discontinuities (see above), these values are consistent with tissue having the larger neurons. It is important to realize that each MADI voxel contains 10^4–10^6 cells (see below), and thus both small glia and large neurons. Therefore, the histogram bars (each from many voxels) measure averaged V values representing glia and neurons.

Although our MADI library ensemble contracted Voronoi cells are computationally sophisticated, they still have relatively simple shapes (figs 1 and 2[12]). One might wonder how the volumes of such cells could ever approximate the Vs of highly articulated neural cell shapes. Some time ago, we considered the possibility that, in cells with such dendritic structures, "the average water molecule lifetime in a process could be significantly smaller than that in a cell body".[45] If this dominated, MADI analyses would discriminate cell processes from cell bodies, and return only the very small V values (substantially less than 1 pL) for these cell parts as estimated above.

However, for the cortex, no significant numbers of pixels are returned with V values below 0.2 pL (200 fL) (Figure 5a). This is true even although there are many library V entries down to 0.0113 pL (11.3 fL) available: the library density is very high at such small V magnitudes. For WM, no significant numbers below 0.4 pL (400 fL) are returned (Figure 5b). On the other end, the upper distribution edges (~11 pL) come nowhere close to the 200 pL library upper limit. This suggests that for by far most of the cells, MADI seems to be effectively integrating the volumes of these very dendritic shapes. Possible exceptions are the largest neurons, whose WM axon projections can reach 10 cm in length. Thus intra-axonal V could be ~20 pL: most axons themselves are found in WM, although their neuronal soma reside in GM. The actual WM intra-axon volume fraction is very small.[46] There are many more smaller oligodendrocytes than axons in WM.[44] These comprise the axon myelin wrapping (see below). Cellularly, WM is dominated by small cells.[28] Almost all intracellular water seems "well-mixed": most water molecules encounter cell membranes many times before they permeate.[12]

How could this be true? There seem two possibilities. First, the permeation probability, p_p, is sufficiently small (typically 0.005[12]) that cell water is physically well mixed. This would mean intracellular water molecules could have to travel hundreds of μm, to visit the entire cell, before permeating. This seems unlikely. Second, intracellular water is in the effectively well-mixed NMR condition. In a different context, we noted the fact a given compartment is physically "well-mixed" does not preclude another compartment "which is in fact not well-mixed" being effectively so.[5] Tissue water can be in an NMR condition without actually being in that physical condition.

In MADI I.,[12] we show how the "well-mixed" NMR condition is established when the nanoscopic diffusion coefficient, D_n, reaches a limiting value in the absence of exchange (the intrinsic D). For large claustrophobia ratios, A/V, this can happen after a t_D of only a few ns. Most tissue extracellular spaces have large A/V values (fig. 2[12], dark blue trajectories), and for dendritic neural cells, the process and somal intracellular spaces also have large A/V values. Thus all brain tissue water enters the well-mixed NMR condition essentially instantaneously. Every water molecule anywhere inside a highly dendritic cell encounters cell membranes very frequently. This is even faster when there is exchange.[12] An important feature of the well-mixed condition is that the intrinsic D_n for intracellular water is zero ($D_{in} \rightarrow 0$); while, for extracellular water, $D_{well\text{-}mixed}$ is nonzero, with a magnitude depending mostly on the tissue V and ρ values.[12] Even in the presence of exchange, the D_{in} and D_{out} quantities limit in distinguishable values.[12] Thus these discriminate intracellular and extracellular water, and allow us to estimate the voxel mean cell volume. There may be no other way to do this in vivo. This seems a very fortunate circumstance for brain MADI. One parameter, V, represents a very parsimonious characterization of the cell: the mathematical description of a dendritic cell shape can require seven parameters.[47]

Measurements by modern fluorescence and optical microscopy on cultured, separated cells find V values from 0.8 to 4 pL for infiltrating lymphoblasts,[48] cancer cells,[49] kidney cells,[50] and other more simply shaped cells.[51]

4.2.2 | Cell number density (ρ)

Independent **ρ** estimates can only come from ex vivo histopathology. Because MRI is inherently a volume-based modality, such **ρ** estimates from classical histology must be determined as stereological (3D from 2D) transformations. However, as detailed above, death, excision, and/or fixation has likely already altered the 2D **ρ** precursors in the ex vivo tissue from their in vivo values pertinent for MADI. Some recent quantitative analyses are based on cell (nucleus) number-counting approaches (reviewed[52]), and are so optimized. The isotropic fractionation (IF) method[28,52–56] is inherently 3D (stereological). These can yield 2D or 3D cell nuclei counting maps. Because death/excision/fixing changes do not alter relative nuclei counts, not all investigators are overly concerned with such procedural aspects: the counting methods are not intended for absolute **ρ** determinations. Furthermore, cell counting methods rely on nuclear staining.[54] Incomplete staining could lead to systematically underevaluated **ρ** estimates: again, not so much of a problem for relative cell counts. However, absolute **ρ** value errors are likely unavoidable. Values for **ρ** can be stereologically (3D from 2D) estimated from classical histopathology results. Examples for **ρ** estimates are machine-learning approaches determining 2D cell density ("cellularity", cells/unit area).[23] But, for example, tissue shrinkage would cause **ρ** overestimation, compared with the in vivo value.

For all of the reasons above, the ex vivo histopathology literature for estimating ρ magnitudes is mixed. All IF studies find ρ_{cGM} is smaller than ρ_{WM}, with values generally in the tens to hundreds of thousands (10^4 to 10^5) cells/μL, with considerable variance.[28,52–56] Some of this is due to tissue heterogeneity: a variation of 75 x 10^3 cells/μL across the ex vivo chimpanzee neocortex has been reported, with a (95% CI) spread of 120 x 10^3 cells/μL for measurements at any location.[53]

We are pleased our primitive model yields in vivo ρ values of similar general magnitudes as estimated from ex vivo histopathology. Table 2 indicates the median in vivo cGM cell density 1.1 x 10^5 cells/μL, while the median WM ρ is 6.9 x 10^5 cells/μL. Conversely to V, ρ values would be much larger if MADI saw soma and neurites as separate.

Whatever the absolute errors, MADI ρ and V maps should be sensitive to the same pathologies (e.g., neuronal degeneration), pharmaceutical responses, and genetic alterations as ex vivo cell nuclei counts,[52] with the significant advantage of being measured in vivo noninvasively.

4.2.3 | Cellular water efflux rate constant (k_{io})

As detailed in the companion paper[12] and in the Introduction, it has been exceedingly difficult (if possible) to accomplish in vivo brain k_{io} measurements. Thus there are few with which to compare MADI measurements, and these are itemized in Table 3.

The fourth row shows Filter-EXchange-Imaging (FEXI) results, obtained from in vivo human brain.[24] These k_{io} values are very small, comparable with those we estimate above for dead cells. The third row shows results from ex vivo brain tissue models. These are roughly an order of magnitude larger. In superfused slices, k_{io} can rise above 3 s^{-1} at a high neuronal firing rate.[7] But ex vivo tissue preparation can cause synaptic disruption,[57] which would reduce metabolic activity. The results in the second row are from anesthetized, living rat brain. These are of similar magnitudes to the ex vivo k_{io} model values. Anesthesia could be expected to reduce metabolic activity. Our (Table 2) results from the awake, resting human brain are given in the first row. The median cGM value (6.6 s^{-1}) is within the anesthetized rat brain value range, but our upper range values are an order of magnitude larger. Our median WM value (22 s^{-1}) is two orders of magnitude larger than the FEXI value.

The rows 2–4 results arise from s-s studies of three different types (SI.IV). We detail in the companion paper[12] the difficulties of achieving a sufficiently large s-s. In particular, the crucial FEXI exchange time generally begins near $t_D = t_v$, the point at which the diffusigraphic s-s vanishes[12]: the intracellular and extracellular H_2O D_n values are almost identical. Small s-s values may be why the rows 2–4 k_{io} values are generally small. Most of our awake brain k_{io} magnitudes are greater than believed accessible with s-s approaches (< 10 s^{-1}).

The 22s^{-1} WM k_{io} magnitude may be of particular importance. Some may find this large k_{io} value unexpected. The concentration of much of WM DWI on the D-tensor anisotropy has led the community to focus on myelinated axons, and the general modeling cornerstone is that diffusion radial to the axon is very much smaller than parallel to the axon.[22,58] Thus the modeling has generally employed ensembles of parallel simple, impermeable cylinders.[58] However, k_{io} is not the rate constant for water exiting a myelinated axon. Crossing the axon cell membrane puts a water molecule into only the myelin sheath. The latter is composed of a jelly roll-like wrapping ("whorls") consisting of many layers of alternating cytoplasmic and interstitial spaces (fig. 8,[1]).[59,60] These myelin coatings comprise oligodendrocyte cell processes.[44] Thus, except at the sparse nodes of Ranvier, there are many bilayer membranes (each with its own k_{io}) to cross before an intra-axon water molecule is free of the myelin-coated axon. The probability (p_p) of crossing each of n membranes in series p_p^n is nil because p_p is small (0.005) and n is a substantial integer (>10) (supporting information in ref. [12]). Therefore, the chances of an individual molecule retaining its frequency label (from PFG-encoding) during this period is also nil.[1] The myelin water layers are exceedingly small, less than 5 nm thick,[59,60] and the oligodendrocytes (as well as the axons) contain many metabolizing mitochondria.[44,46] These factors are consistent with intra-myelin k_{io} being very large, and probably most WM metabolic activity occurs within myelin. MADI may be especially adept at detecting myelin metabolic activity, an important finding. There are many significant WM

TABLE 3 Brain k_{io} estimations

	GM		WM	
	k_{io} (s^{-1})	range[a]	k_{io} (s^{-1})	range[a]
awake, resting human	6.6[a–c]	3–72	22[a,b]	12-42
anesthetized rat	1.6,[84] 1.8,[85] 8.6[86]			
ex vivo models	1.3, 1.8,[87] 2.0[6,7]			
awake, resting human	0.21[d,24]		0.26[d,24]	

Abbreviations: FEXI, Filter-EXchange-Imaging; GM, gray matter; WM, white matter.
[a]heterogeneous (this work).
[b]median (Table 2).
[c]cortex.
[d]FEXI (k_{io} = AXR(1 − ρV)/(1 − f_M))[1,83] (ρV = 0.8, f_M[88]).

pathologies.[5,13,14] Ensembles of simple, impermeable cylinders ("sticks") are wonderful models[18,58] underlying elegant WM tractographic applications, but not good models for trans-membrane water exchange and metabolic activity. Such considerations highlight DTI/MADI complementarity.

The PET correlation (Figure 6) does not validate our k_{io} absolute values, but it does support the hypothesis k_{io} is sensitive to metabolic activity, and could be reduced in anesthesia and in some pathologies. Because the V and $\boldsymbol{\rho}$ values show reasonable accuracy (minimal systematic error) and k_{io} is the only other parameter, it might be true by extension that the MADI k_{io} values are also rather accurate.

4.2.4 | Neurogliovascular unit (NGVU) coupling

The interplay between k_{io} and k_{po} (the capillary water efflux [unidirectional] rate constant determined using CA[5]) may also support k_{io} accuracy. The $\langle k_{po}/k_{io}\rangle$ ratio can be considered a quantitative measure of the well-known NGVU metabolic coupling. Thus its smaller value in WM, 0.15 (= 3.2/22), than in cGM, 0.44 (= 2.9/6.6), could represent poorer WM coupling. The WM blood volume fraction (v_b) is ~0.5 that of cGM.[5] But $v_b = \boldsymbol{\rho}^{\dagger}V^{\dagger}$ (the capillary density•mean capillary volume product), and it is likely capillary volumes are similar. If $\boldsymbol{\rho}_{WM}$ is greater than $\boldsymbol{\rho}_{GM}$ (Figure 4b) while $\boldsymbol{\rho}^{\dagger}_{WM}$ is less than $\boldsymbol{\rho}^{\dagger}_{GM}$, there are fewer capillaries/cell in WM than in GM. Thus the average WM cell is further from capillaries, and this could result in poorer NGVU metabolic coupling. The similarities of the WM/GM ratios for $[k_{po}/k_{io}]_{WM}/[k_{po}/k_{io}]_{GM}$ (0.15/0.44) = 0.34, and $[v_b]_{WM}/[v_b]_{GM}$ (0.014/0.031) = 0.45, are consistent with this conjecture. The absolute values of $[k_{po}]_{WM}$, $[k_{po}]_{GM}$, $[v_b]_{WM}$, and $[v_b]_{GM}$ have been independently validated.[5,13–16] NGVU metabolic coupling could be mediated in part by cellular competition for the shared NKA substrate K_o^+ (Figure 1).[10] The fact that WM has a greater myelin content could also contribute.

Because $\langle k_{io}/k_{po}\rangle_{WM}$ is 6.9 (= 22/3.2), brain WM k_{io} maps might exhibit analogous changes as k_{po}[5,13–16] with greater sensitivity: furthermore, no CA is necessary. For example, the WM k_{io} "hot spots" (yellow, Figure 4d) may correspond to WM k_{po} hot spots.[5] Also, WM k_{io} maps could settle the question of whether k_{po} decreases in MS lesions and GBM tumors because of only NGVU coupling disruptions (perhaps by angiogenesis in the tumor) or actual k_{io} decreases.

4.3 | Brain metabolic activity context

Much of what is known about intermediary metabolic pathways comes from purified solution studies. However, quantitative imaging and spectroscopic studies provide information on human brain pathways in the in vivo condition. In Figure 6a, ^{18}FDG-PET gives the tissue metabolic rate of glucose consumption (strictly, uptake) ($^{t}MR_{gl}$) to be 7.9 and 3.6 (pmol [gl_i] consumed)/s/μL (tiss) for cGM and WM, respectively. The main glucose biochemical role is ATP production. ^{31}P-MRS-MT studies yield the metabolic rate of this $^{t}MR_{ATP}$ to be 160 and 50 (pmol [ATP_i] produced)/s/μL (tiss) for cGM and WM, respectively.[61] The divisions 160/7.9 and 50/3.6 give stoichiometric ratios of 20 and 14 ATP_i/gl_i for cGM and WM, respectively. These suggest parallel operation of the oxidative phosphorylation (36 ATP_i/gl_i) and glycolytic (2 ATP_i/gl_i) pathways.[1] ^{13}C-MRS and [^{14}C]-2-deoxyglucose autoradiography studies have yielded refined "energy budgets" for human brain WM and cGM.[56] These assay ATP_i usage by biochemical task, not by particular enzyme flux. (NKA is surely used for a number of different tasks.) Thus they determined the tissue metabolic rates of all ATP_i consumption (the total of all ATPase activities, $^{t}MR_{ATPases}$) to be 170 and 110 (pmol [ATP_i] consumed)/s/μL (tissue) for cGM and WM, respectively.[56] It is satisfying that, for cGM, these two very different methods give almost identical values for ATP_i production $^{t}MR_{ATP}$, 160,[61] and consumption $^{t}MR_{ATPases}$, 170 pmol (ATP_i)/s/μL (tiss)[56]: cGM ATP_i is homeostatic. Why these values would be different for WM, 50 and 110 pmol (ATP_i)/s/μL(tiss) is not clear: WM ATP_i is also homeostatic. It might have to do with glycolytic pathways.[62] However, both oxphos and aerobic glycolysis are detected by ^{31}P-MRS-MT in the heart.[63] In any case, all results show that MR_{cGM} is greater than MR_{WM}.

Our measurements yield cellular metabolic rates (^{c}MR). Thus the Figure 6a ordinate cGM and WM ($0.042 \bullet k_{io} \bullet V = x^{c}MR_{NKA}$) values are 1.0 and 0.83 nmol(H_2O_i)/s/cell, respectively (Table 2). To compare with literature, we must divide reported tissue MR measures by the Table 2 $\boldsymbol{\rho}$ values. This is a virtue of determining $\boldsymbol{\rho}$. For cGM, this gives (170 pmol (ATP_i)/s/μL (tiss))/(1.1 x 10^5 cells/μL (tiss)) = 1.5 fmol (ATP_i) consumed/s/cell. For WM, this gives (110 pmol (ATP_i)/s/μL (tiss))/(6.9 x 10^5 cells/μL (tiss)) = 0.16 fmol (ATP_i) consumed/s/cell. These are the cellular values for total ATP_i consumed per second, that is, by all ATPases. We can estimate lower limits for the stoichiometric ratio x, by treating all ATP_i as if consumed by NKA. Thus we obtain for cGM, x ≥ 0.67 x 10^6 H_2O_i/ATP_i ((1.0 nmol [H_2O_i]/s/cell)/(1.5 x 10^{-6} nmol [ATP_i] /s/cell)). For WM, we obtain, x ≥ 5.2 x 10^6 H_2O_i/ATP_i ((0.83 nmol [H_2O_i]/s/cell)/(1.6 x 10^{-7} nmol [ATP_i]/s/cell)). These x estimates are quite consistent with the Figure 6 quantitative aspects (assuming consistent glucose conversion to ATP). They represent lower limits because, although NKA is often the major ATP consumer, there are a number of other ATPases. Such large stoichiometries suggest aquaporin participation in active trans-membrane water cycling (Figure 1). The literature x' values for individual transporters (10–10^3) were measured in ex vivo models,[9] which could be a significant fact. If cGM and WM do actually have different x values (Figure 6a), this is still a metabolic pathway change; a difference in AWC water cotransporters, glucose conversion to ATP, or both. As mentioned above, individual WM neuronal tracts are not as neuron-rich as cGM, and are known to transmit signals between multiple cGM regions.[28,64,65]

The $\langle k_{io}\rangle_{WM}/\langle k_{io}\rangle_{GM}$ ratio (~3.3) may represent a significant, fruitful, finding. Other than V differences, what else could cause the WM k_{io} value to be larger than in GM? It is known that WM water diffusion is considerably more anisotropic than in GM. However, as detailed in,[12] our model and acquisition is focused on the DT magnitude, not its anisotropy. Furthermore, the MADI biomarkers (k_{io}, $\boldsymbol{\rho}$, V) are not formally anisotropic by definition. Still, it might be possible some anisotropy effect is manifest due to experimental imperfection.

4.4 | Future improvements

Because our method is numerical, we have begun assembling canonical simulation "libraries" ("dictionaries"): each entry is "labeled" with its three-parameter (k_{io}, ρ, V) "address", and each library is universal: there is only one of its kind. Every tissue and biological condition could have a unique DWI signature ("fingerprint"). Our first library stores water molecule displacement curves, $d^{rms} = f\,(t_{RW})$.[12] A second library stores simulated b-space decays for SDE-CT acquisitions (Figures 2 and 3). As we have seen (Figure 5, Table 2), the major MADI challenge is finding a unique library decay to best match any tissue experimental decay (the inverse problem).

For example, as is clear from the above, we cannot tell whether the greater GM heterogeneity seen in all MADI parametric maps (Figures 4-6, Table 2), compared with WM, reflects the well-known greater inherent GM tissue cellular heterogeneity[28,30] or the SDE-CT library granularity. Figure 6c shows a recent quantitative literature ^{18}FDG-PET map, with modern resolution. Nonetheless, its still poor resolution causes an apparent smoothing of the GM regions. Thus PET is also not resolving GM heterogeneity. Perhaps improved MADI precision will be able to settle this issue. There are two general ways to proceed: (1) improve the libraries; (2) improve the DWI acquisitions. Both of these avenues should be followed.

4.4.1 | Filling in the libraries

Of necessity, the libraries contain discrete parameter entries. These were being populated at the same time the correct in vivo values of the novel V, $\boldsymbol{\rho}$, and k_{io} parameters were being established by comparison of results in Figures 4-6 and Tables 2 and 3 with careful analyses of pertinent in vitro and ex vivo literature, as well as consideration of NMR principles. It is important that decays appropriate for different tissues are well represented in the libraries. Furthermore, the analysis of the test results in Figure 5 and Table 2 reveal the spacing of these decays has to be quite fine-grained. This great precision sensitivity to library entry spacings was not known beforehand. Thus library densities must be increased in appropriate tissue regions.

A current deficiency is the presence of only 20 different v_i ($= \boldsymbol{\rho}$•V) values. Because entries can lie on only the 1300 different $\boldsymbol{\rho}$•V hyperbolae (65 k_{io} values x 20 v_i values), this is a significant limitation. For the brain, this shows up particularly in GM parameter uncertainties (Figures 4-6, Tables 2 and 3). GM water diffusivity is generally greater than that of WM water (Figure S1b), despite the fact that R_2(GM) is smaller than R_2(WM).[12] Thus the WM region is higher in b-space (Figure 3) than the GM region; an example of only the latter is shown in Figure 3. It is very clear we need more library entries in the GM region; particularly filling in not only the sparse v_i values, but the sparse parameter values at larger parameter magnitudes. We would hope for an ultimate precision for each parameter approaching half its library step size. It is highly likely parameter estimations will become more precise as the library becomes more dense.

Our model parameters approach the expected limiting values: $k_{io} \to +\infty$ (Figure 2d,g), $\rho \to 0$ (Figure 2e,h), and $V \to 0$ (Figure 2f,i), for pure water. This could make MADI maps quite sensitive to regions with significant pools of acellular saline, via partial volume averaging. Possible examples are brain ventricular, subarachnoid, perivascular (Virchow–Robin) spaces, deliberately omitted in Figures 4 and 6. However, to avoid artifactually incorrect parameter values (particularly of k_{io}), both $\boldsymbol{\rho}$ and V must approach zero together, that is, v_i must $\to 0$. Our current SDE-CT decay library does not have any entries for which both $\boldsymbol{\rho}$ and V are simultaneously very small (but finite) in the range $0 < v_i < 0.5$. Filling in such entries could provide the possibility for sensitive mapping of secretory tissue water-rich regions (e.g., glandular ductule lumens in pancreas, prostate, breast).

4.4.2 | Improving DWI acquisitions

Obviously, increased experimental S/N should increase MADI parameter precisions (the Figure 5, Table 2 computational tests used S/N = 50).

4.4.3 | Higher magnetic field strength

One can increase S/N by increasing $\mathbf{B}_0$. The data in this paper were obtained at $\mathbf{B}_0 = 3$ T: there are now human instruments with $\mathbf{B}_0 = 10.5$ T. To first approximation, b is normalized for $\mathbf{B}_0$ and T_2 (SI.II.).

4.4.4 | Higher b-space

As is obvious in Figure 3, the noiseless library SDE decays diverge from each other at large b values: they all start at the same location, b = 0. Thus the chance of discriminating entries increases at high b. Unfortunately, however, that is also where the S/N is poorest (also Figure 3).

4.4.5 | Improved RF transceiver coils

Improved RF transceiver coils canincrease S/N. The Human Connectome Project (HCP) efforts combine resting-state functional (T_2* signal fluctuations) with DWI structural (tractography) data. Thus HCP hardware and acquisition advances have enabled significant DWI quality and speed improvement.[64] For example, a record in vivo, resting human brain b_{max} value, near 40 ms/μm^2 (40,000 s/mm^2), has been (under) reported.[66] These results (Figure S1b) indicate there can be still meaningful brain signal (10% to 3% of S_0) at b = 10 ms/μm^2 (10,000 s/mm^2). Of course, there is even more signal at smaller b values: for example, at b = 7000 s/mm^2, which should be accessible with current clinical SDE-CT, there is ~20% and ~6% of S_0 for WM and GM, respectively. Even larger b values might be clinically accessible with SDE-CG (incrementing t_D) acquisitions (SI.I.). Our SDE decay simulations are artifact-free down to S = 0.015 S_0, and a bit lower.

4.4.6 | Improved k-space navigation

With novel k-space sampling trajectories, the speed of DWI acquisition can be increased.[67] Thus, keeping the total imaging time (TT) constant, one can increase signal averaging, and thus S/N.

4.4.7 | Data denoising

There are tissues that naturally yield experimental decays with relatively poor S/N (Figure 3). Such data can be numerically denoised.[68,69]

4.4.8 | Weighted b acquisitions

One might want to disproportionately weight the number of b measurements limited by a clinically acceptable TT. For example, if high b values are desirable, one might want a larger number of such.

4.4.9 | Articulated b-space decays

Unfortunately, as noted above, SDE b-space decays are relatively featureless (Figures 2 and 3). However, there are acquisitions that can "articulate" these, that is, intentionally add decay features. Adding a deliberate feature ("wrinkle") to the b-space decay should make it more distinctive. An example is an interrupted double diffusion encoding approach.[70] The magnetization is transiently stored against transverse dephasing, while molecular diffusion and exchange continue. This adds a deliberate discontinuity that is particularly sensitive to exchange.[70] Such could render different tissue signatures more distinguishable. At sufficiently large b, an SDE-CG acquisition provides a natural articulation (SI.I.).

Also, it has been shown that different acquisitions can elicit different b-space decays from the same tissue. An example is SDE and oscillating gradient spin-echo encoding.[23,71] Although SDE-CT and SDE-CG decays are essentially identical at low b values (SI.I.), they diverge at higher b.[56,72,73]

Employing our displacement (d^{ms}) curve library,[12] which costs most of the computation time, we can retrospectively apply any diffusion-encoding acquisition pulse sequence[23,74,75] to calculate its b-space decay for any k_{io},$\boldsymbol{\rho}$,V combination (eqn. 8[12]). This "richness" could add considerably to mitigating the inverse problem, and reducing parameter uncertainties. Thus, future MADI DWI libraries must be labeled with their acquisition particulars (e.g., SDE-CT, SDE-CG, OGSE, etc.; δ, G, Δ, etc).

4.4.10 | Noise caused by tissue motion

Some bodily tissues are more subject to motion (of cardiac, respiratory, etc. origins) than others. There are elegant acquisitions designed to decrease such sensitivity (for heart, pancreas, breast, etc.).[76–78]

4.4.11 | Blood flow effects

There is concern blood flow affects the SDE decay at very small b values: the intra-voxel incoherent motion (IVIM) mechanism.[79] There is a DWI acquisition designed to reduce sensitivity to IVIM.[80]

4.4.12 | Parameter anisotropy

It will be interesting to determine if the aMRI parameters (k_{io}, $\boldsymbol{\rho}$, V) exhibit any anisotropy. By formal definition, they should not.

5 | CONCLUSIONS

Our results show metabolic activity information can be extracted, and mapped, from noninvasive, high-resolution in vivo DWI signals. Results are presented for the awake, resting human brain. The WM and GM metabolic activities are consistent with ^{18}FDG-PET and with predictions from in vitro and ex vivo model studies. Furthermore, we find the metabolic pathways utilized in WM in vivo are likely different from those in GM.

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CONFLICT OF INTEREST

CSS, EMB, XL, BM, GJW, WDR, and JHM are inventors on U.S. Provisional.

Patent Application (No. U.S. 62/482,520) "*Activity MRI*."

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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